AWS VPC PEERING PROJECT

Private Connectivity Between Two Amazon VPCs

Project Report

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Batch: AWS Internship batch 1

AWS Region: Asia Pacific (Mumbai) — ap-south-1

Completion date: 16/10/26

Prepared for the EWB Courses AWS Networking / AWS Architect Associate student project.

1. Abstract

This project demonstrates private communication between workloads placed in two separate Amazon Virtual Private Clouds (VPCs) using an AWS VPC peering connection. VPC A contains a public subnet for a bastion host and a private subnet for App-A. VPC B contains a private subnet for App-B. Route tables on both sides provide forward and return paths, while security groups restrict SSH and ICMP traffic to specific source addresses. App-B is accessed from the laptop through the bastion host using SSH ProxyJump; the private application instances are not assigned public IPv4 addresses.

2. Objectives

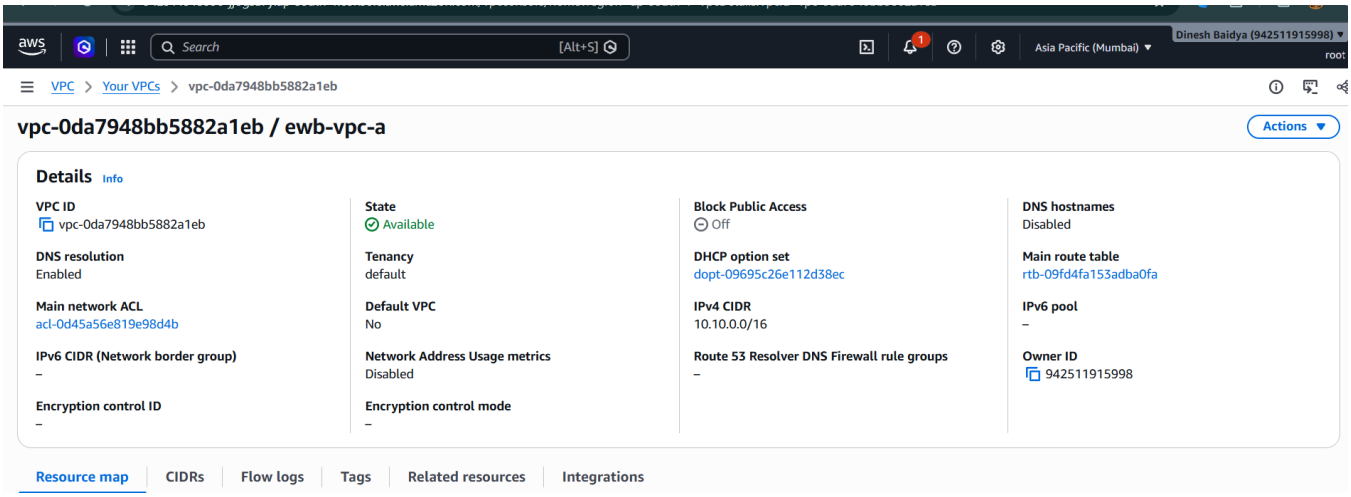
- Create two non-overlapping VPCs and the required public/private subnets.
- Connect the VPCs using a VPC peering connection.
- Configure the correct routes in both directions and retain the internet-gateway default route for the bastion subnet.
- Restrict access with security-group rules using specific source IPs.
- Verify two-way ICMP connectivity and SSH access to App-B through the bastion host.
- Perform a controlled ICMP failure-and-recovery test and document cleanup.

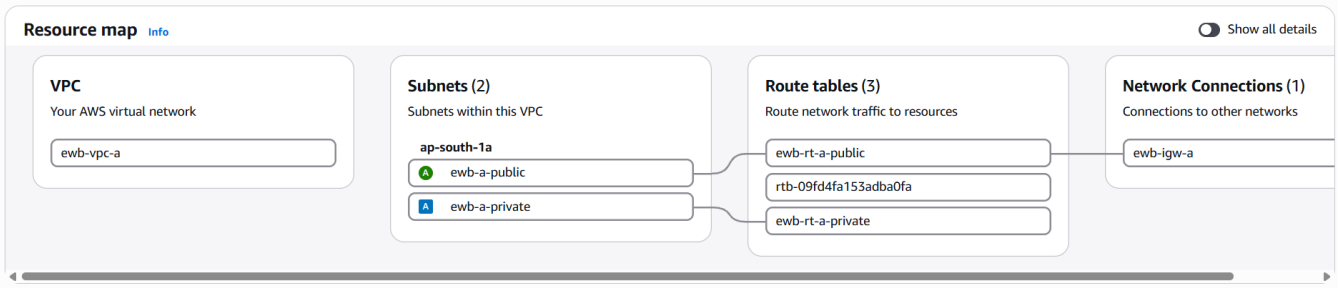
3. Architecture and Addressing

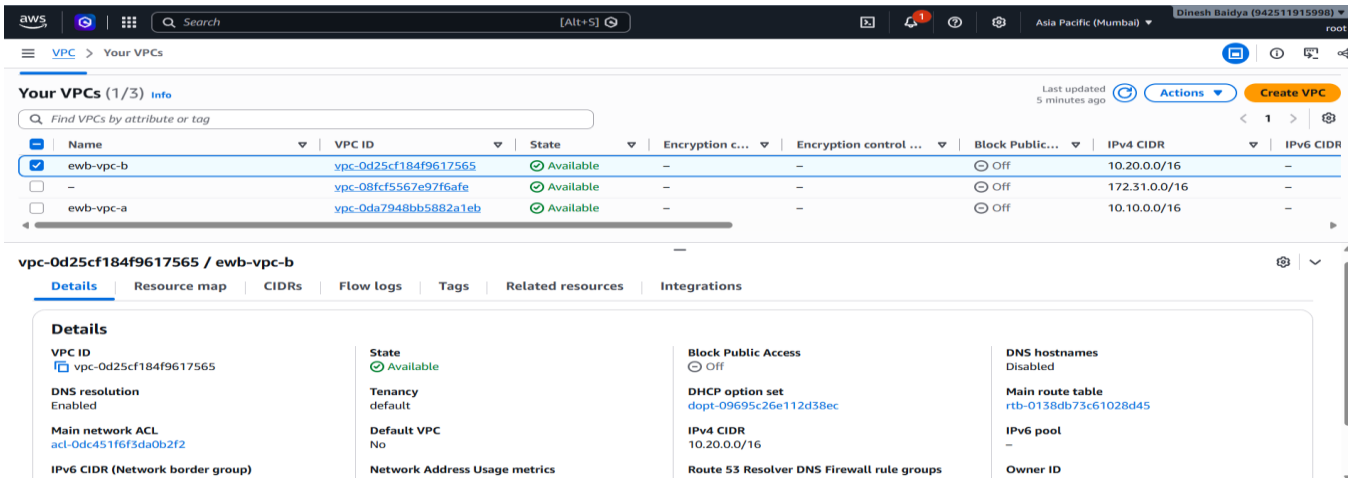
The topology below is a text summary. Replace or supplement it with a clear architecture diagram showing the actual resource IDs, subnet CIDRs, peering connection, route directions and instance private IPs.

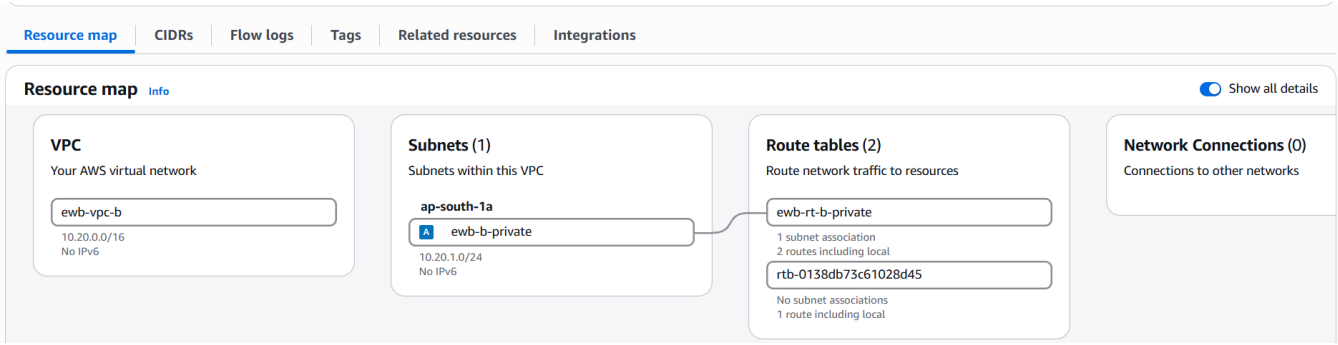
Component	CIDR / IP	Purpose / details
VPC A	10.10.0.0/16	Hosts the bastion and App-A
VPC A public subnet: ewb-a-public	10.10.1.0/24	Bastion subnet; route to Internet Gateway and peering connection
VPC A private subnet: ewb-a-private	10.10.2.0/24	Private workload App-A
VPC B	10.20.0.0/16	Hosts private workload App-B
VPC B private subnet: ewb-b-private	10.20.1.0/24	Private workload App-B
Bastion: ewb-bastion	Public IP 35.154.170.65; private IP 10.10.1.39	SSH entry point
App-A: ewb-app-a	Private IP 10.10.2.192	Private instance in VPC A; no public IPv4
App-B: ewb-app-b	Private IP 10.20.1.96	Private instance in VPC B; no public IPv4
VPC peering connection	pcx-0e2a0f1d635f24a36	Private routing between VPC A and VPC B

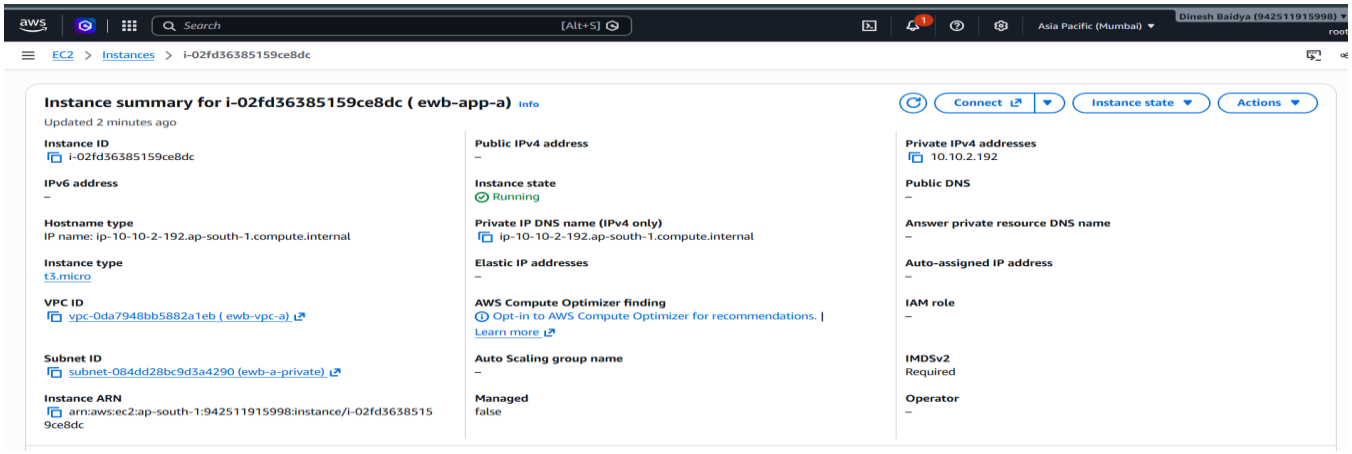
Architecture diagram / VPC resource overview

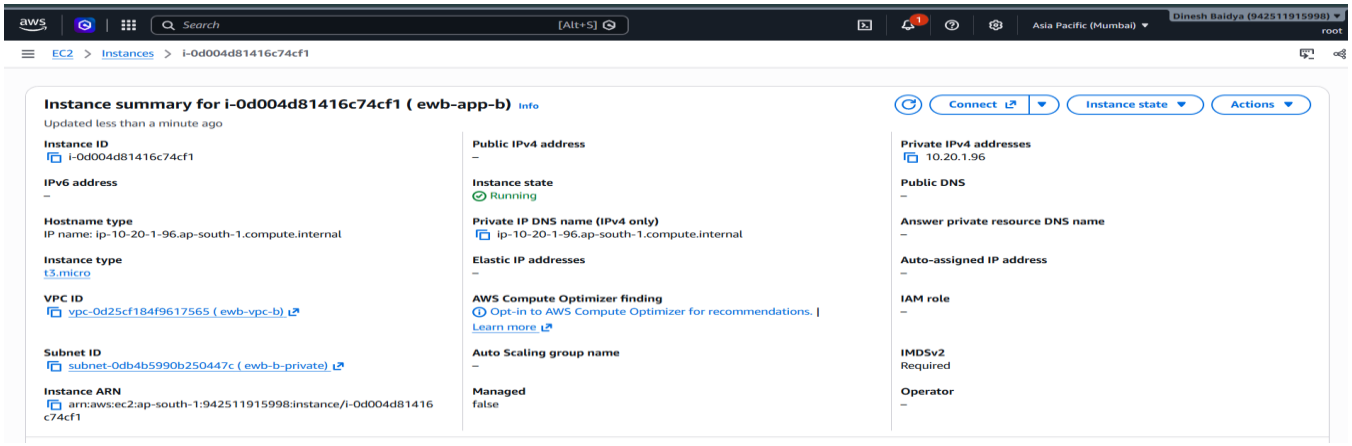


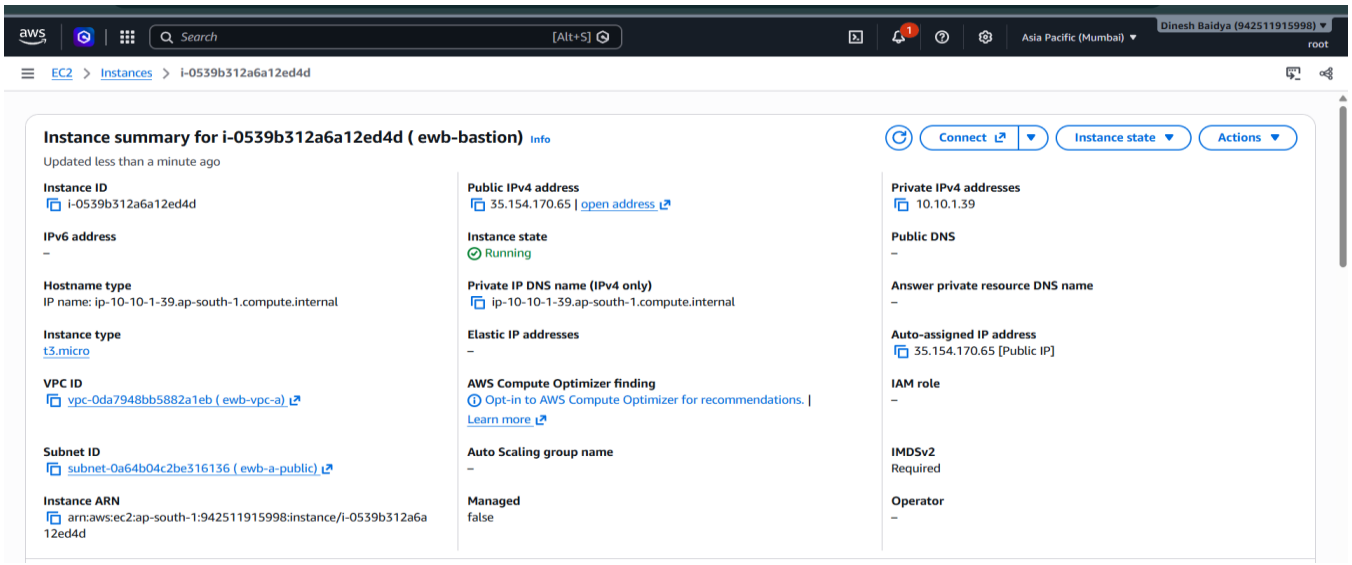

Resource map Info
Show all details
VPC
Your AWS virtual network
ewb-vpc-a
Subnets (2)
Subnets within this VPC
ap-south-1a
ewb-a-public
ewb-a-private
Route tables (3)
Route network traffic to resources
ewb-rt-a-public
rtb-09fd4fa153adba0fa
ewb-rt-a-private
Network Connections (1)
Connections to other networks
ewb-igw-a

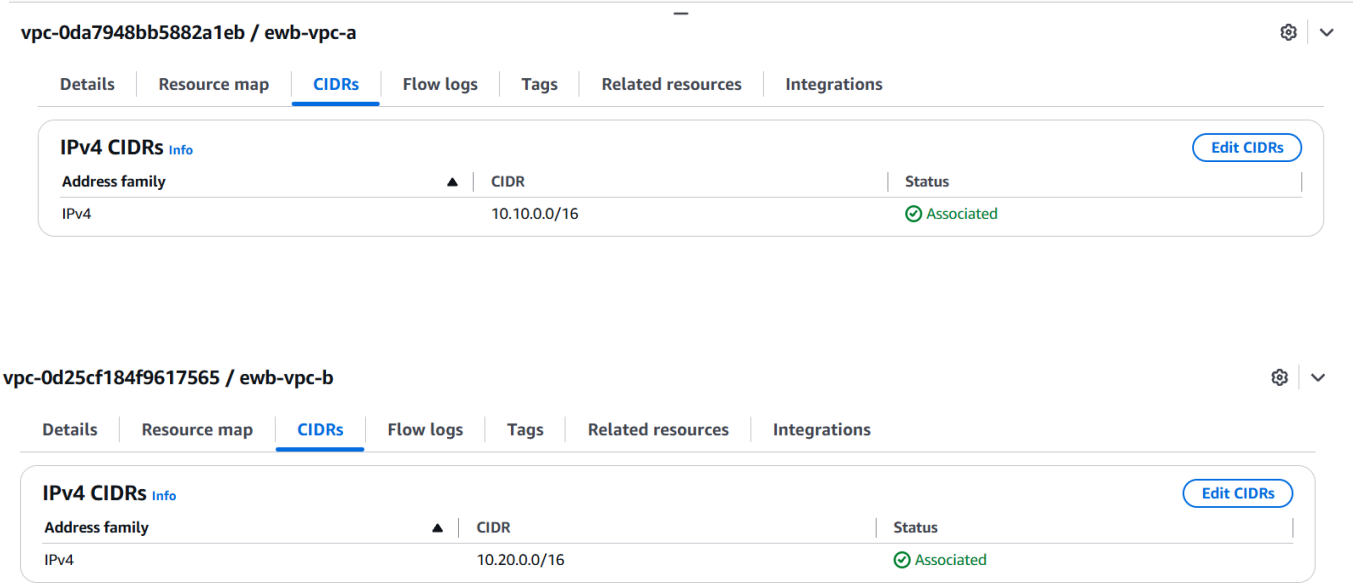

Your VPCs (1/3) Info
Actions
Create VPC
Name
VPC ID
State
ewb-vpc-b
vpc-0d25cf184f9617565
Available
10.20.0.0/16
vpc-08fcf5567e97f6afe
172.31.0.0/16
ewb-vpc-a
vpc-0da7948bb5882a1eb
10.10.0.0/16
vpc-0d25cf184f9617565 / ewb-vpc-b
Details
Resource map
CIDRs
Flow logs
Tags
Related resources
Integrations
VPC ID
vpc-0d25cf184f9617565
State
Available
Block Public Access
Off
DNS hostnames
Disabled
DNS resolution
Enabled
Tenancy
default
DHCP option set
dopt-09695c26e112d38ec
Main route table
rtb-0138db73c61028d45
Main network ACL
acl-0dc451f6f3da0b2f2
Default VPC
No
IPv4 CIDR
10.20.0.0/16
IPv6 pool
IPv6 CIDR (Network border group)
Network Address Usage metrics
Route 53 Resolver DNS Firewall rule groups
Owner ID


Resource map
CIDRs
Flow logs
Tags
Related resources
Integrations
Resource map Info
Show all details
VPC
Your AWS virtual network
ewb-vpc-b
10.20.0.0/16
No IPv6
Subnets (1)
Subnets within this VPC
ap-south-1a
ewb-b-private
10.20.1.0/24
No IPv6
Route tables (2)
Route network traffic to resources
ewb-rt-b-private
1 subnet association
2 routes including local
rtb-0138db73c61028d45
No subnet associations
1 route including local
Network Connections (0)
Connections to other networks


EC2 > Instances > i-02fd36385159ce8dc
Instance summary for i-02fd36385159ce8dc (ewb-app-a) Info
Updated 2 minutes ago
Connect
Instance state
Actions
Instance ID
i-02fd36385159ce8dc
Public IPv4 address
Private IPv4 addresses
10.10.2.192
IPv6 address
Instance state
Running
Public DNS
Hostname type
IP name: ip-10-10-2-192.ap-south-1.compute.internal
Private IP DNS name (IPv4 only)
ip-10-10-2-192.ap-south-1.compute.internal
Answer private resource DNS name
Instance type
t3.micro
Elastic IP addresses
Auto-assigned IP address
VPC ID
vpc-0da7948bb5882a1eb (ewb-vpc-a)
AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations.
Learn more
IAM role
Subnet ID
subnet-084dd28bc9d3a4290 (ewb-a-private)
Auto Scaling group name
IMDSv2
Required
Instance ARN
arn:aws:ec2:ap-south-1:942511915998:instance/i-02fd3638515
9ce8dc
Managed
false
Operator


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Dinesh Baidya (942511915998)
root
EC2 > Instances > i-0d004d81416c74cf1
Instance summary for i-0d004d81416c74cf1 (ewb-app-b) Info
Updated less than a minute ago
Connect
Instance state
Actions
Instance ID
i-0d004d81416c74cf1
IPv6 address
–
Hostname type
IP name: ip-10-20-1-96.ap-south-1.compute.internal
Instance type
t3.micro
VPC ID
vpc-0d25cf184f9617565 (ewb-vpc-b)
Subnet ID
subnet-0db4b5990b250447c (ewb-b-private)
Instance ARN
arn:aws:ec2:ap-south-1:942511915998:instance/i-0d004d81416c74cf1
Public IPv4 address
–
Instance state
Running
Private IP DNS name (IPv4 only)
ip-10-20-1-96.ap-south-1.compute.internal
Elastic IP addresses
–
AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations.
Learn more
Auto Scaling group name
–
Managed
false
Private IPv4 addresses
10.20.1.96
Public DNS
–
Answer private resource DNS name
–
Auto-assigned IP address
–
IAM role
–
IMDSv2
Required
Operator
–


Search
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Dinesh Baidya (942511915998)
root
EC2 > Instances > i-0539b312a6a12ed4d
Instance summary for i-0539b312a6a12ed4d (ewb-bastion) Info
Updated less than a minute ago
Connect
Instance state
Actions
Instance ID
i-0539b312a6a12ed4d
IPv6 address
–
Hostname type
IP name: ip-10-10-1-39.ap-south-1.compute.internal
Instance type
t3.micro
VPC ID
vpc-0da7948bb5882a1eb (ewb-vpc-a)
Subnet ID
subnet-0a64b04c2be316136 (ewb-a-public)
Instance ARN
arn:aws:ec2:ap-south-1:942511915998:instance/i-0539b312a6a12ed4d
Public IPv4 address
35.154.170.65 | open address
Instance state
Running
Private IP DNS name (IPv4 only)
ip-10-10-1-39.ap-south-1.compute.internal
Elastic IP addresses
–
AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations.
Learn more
Auto Scaling group name
–
Managed
false
Private IPv4 addresses
10.10.1.39
Public DNS
–
Answer private resource DNS name
–
Auto-assigned IP address
35.154.170.65 [Public IP]
IAM role
–
IMDSv2
Required
Operator
–


vpc-0da7948bb5882a1eb / ewb-vpc-a
Details
Resource map
CIDRs
Flow logs
Tags
Related resources
Integrations
IPv4 CIDRs Info
Edit CIDRs
Address family
CIDR
Status
IPv4
10.10.0.0/16
Associated

vpc-0d25cf184f9617565 / ewb-vpc-b
Details
Resource map
CIDRs
Flow logs
Tags
Related resources
Integrations
IPv4 CIDRs Info
Edit CIDRs
Address family
CIDR
Status
IPv4
10.20.0.0/16
Associated

4. Resource Inventory

The values below include identifiers available from the lab session. Fill in the remaining IDs from the AWS console before submission.

Resource	Name / ID	Notes
VPC A	Name: wb-vpc-a | VPC ID: vpc-0da7948bb5882a1eb	CIDR 10.10.0.0/16
VPC B	Name: wb-vpc-b | VPC ID: vpc-0d25cf184f9617565	CIDR 10.20.0.0/16
Peering	pcx-0e2a0f1d635f24a36	Active
Bastion instance	i-0539b312a6a12ed4d	ewb-bastion; 35.154.170.65 / 10.10.1.39
App-A instance	i-02fd36385159ce8dc (ewb-app-a)	ewb-app-a; 10.10.2.192; no public IPv4
App-B instance	i-0d004d81416c74cf1 (ewb-app-b)	ewb-app-b; 10.20.1.96; no public IPv4
App-B subnet	subnet-0db4b5990b250447c	10.20.1.0/24
App-B security group	sg-018b7ee83cb375164	ewb-sg-app-b
VPC A public route table	rtb-01e3c5a7d386325a0 / ewb-rt-a-public	ewb-rt-a-public
VPC A private route table	rtb-0052937b82db323eb / ewb-rt-a-private	ewb-rt-a-private
VPC B private route table	rtb-0a5444b03c5c581a0 / ewb-rt-b-private	ewb-rt-b-private
Internet Gateway	igw-05acc7673e2ab44e6 / ewb-igw-a	Attached to VPC A
App-A security group	sg-0331abf925cc5e144 - ewb-sg-app-a	Record actual ID and inbound rules
Bastion security group	sg-018b7ee83cb375164 - ewb-sg-app-b	SSH inbound restricted to current public IP /32

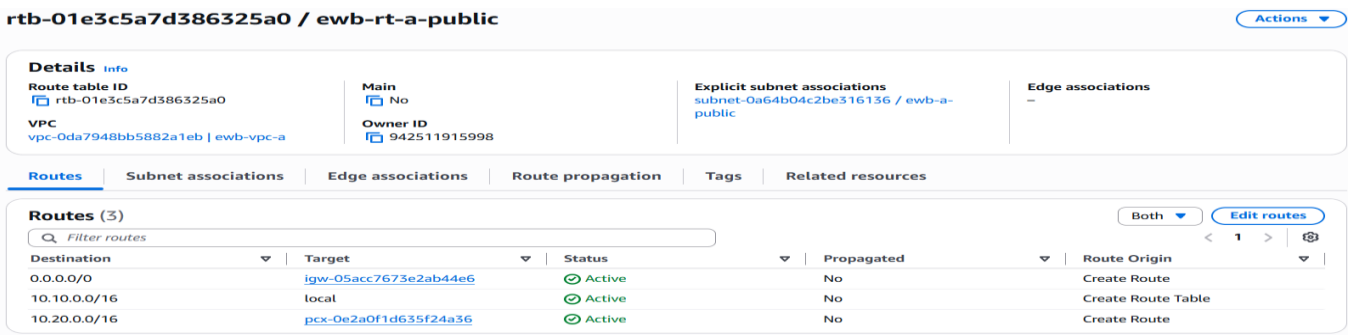
5. Routing Configuration

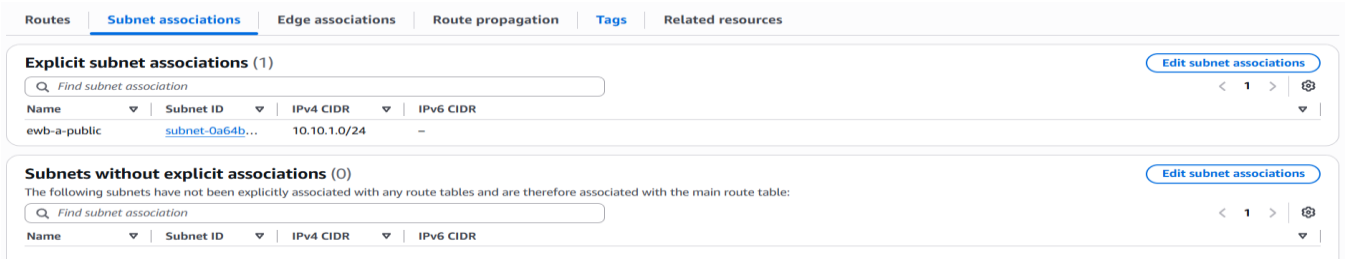
AWS automatically creates a local route for each VPC CIDR. The additional routes below provide peering reachability. The public route table must retain its default route to the Internet Gateway; the more-specific VPC B route sends VPC B traffic over peering.

Route table	Destination	Target	Association / purpose
ewb-rt-a-public	10.10.0.0/16	local	ewb-a-public; VPC A local traffic
ewb-rt-a-public	0.0.0.0/0	ewb-igw-a (actual IGW ID)	Bastion internet access
ewb-rt-a-public	10.20.0.0/16	pcx-0e2a0f1d635f24a36	Bastion-to-App-B route through peering
ewb-rt-a-private	10.10.0.0/16	local	ewb-a-private; VPC A local traffic
ewb-rt-a-private	10.20.0.0/16	pcx-0e2a0f1d635f24a36	App-A to VPC B
ewb-rt-b-private	10.20.0.0/16	local	ewb-b-private; VPC B local traffic
ewb-rt-b-private	10.10.0.0/16	pcx-0e2a0f1d635f24a36	Return path from App-B to VPC A

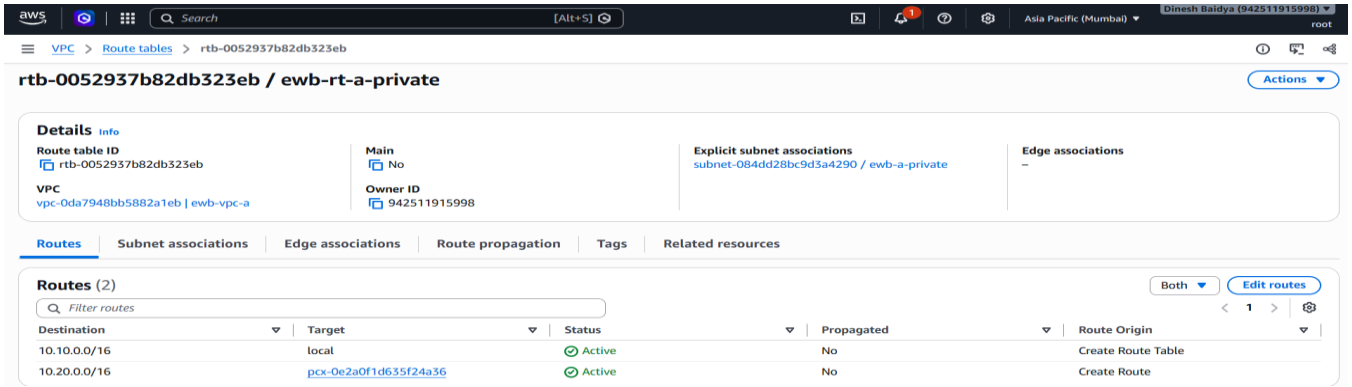
Important: capture the subnet association view as well as the route list for all three route tables.

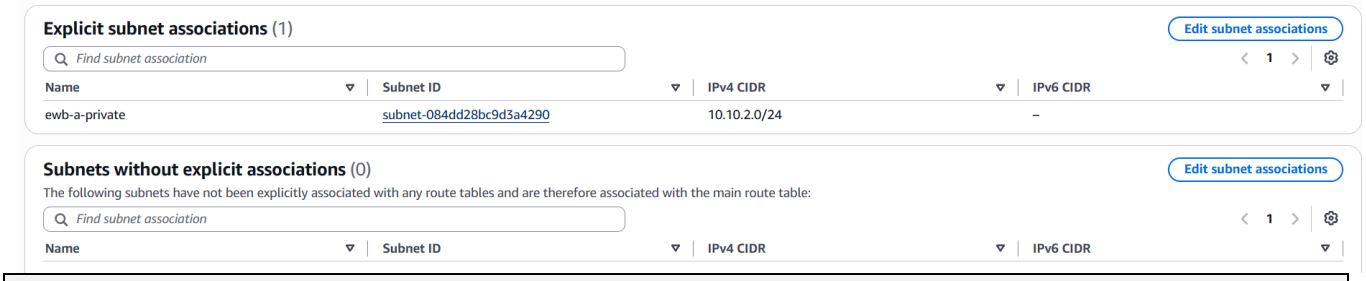
ewb-rt-a-public routes and subnet association





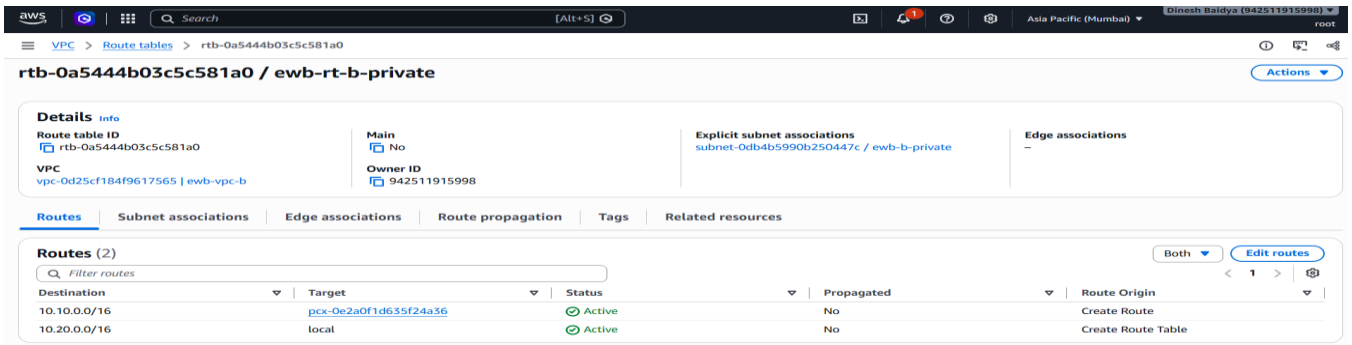
ewb-rt-a-private routes and subnet association

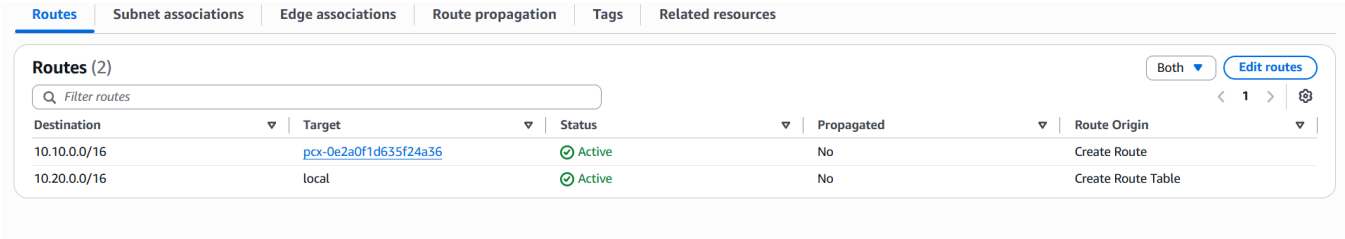




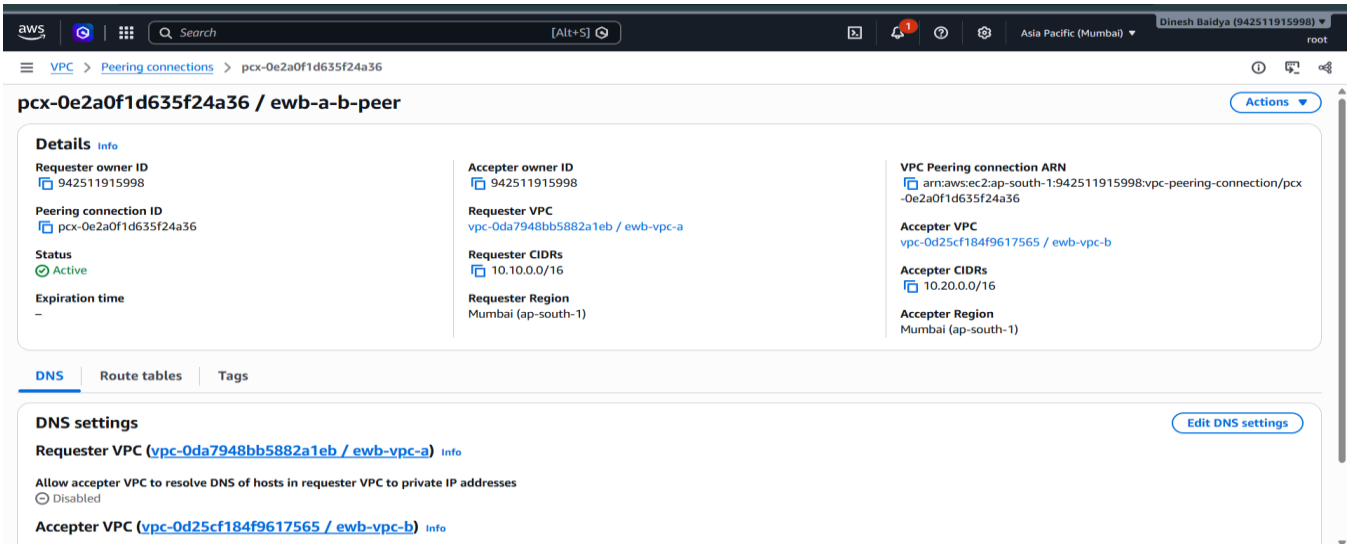
ewb-rt-b-private routes and subnet association

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VPC peering connection showing its actual state

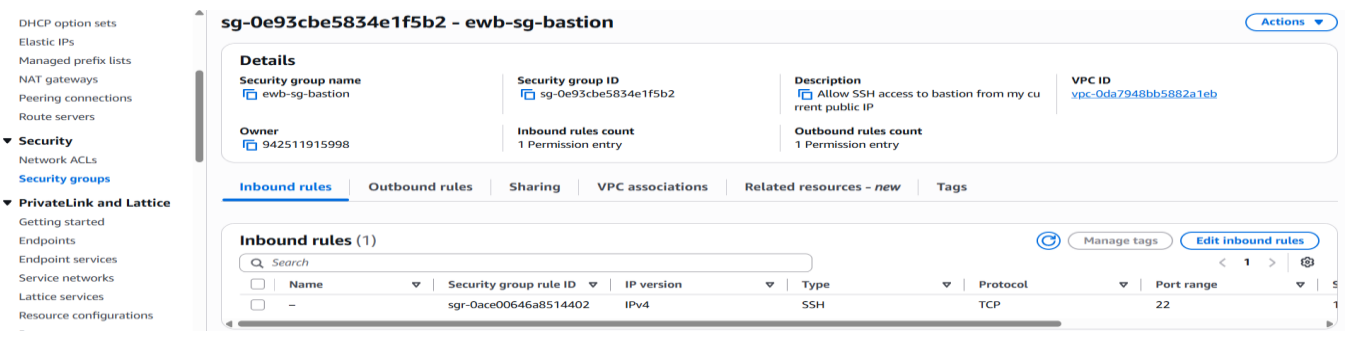


6. Security Group Rules

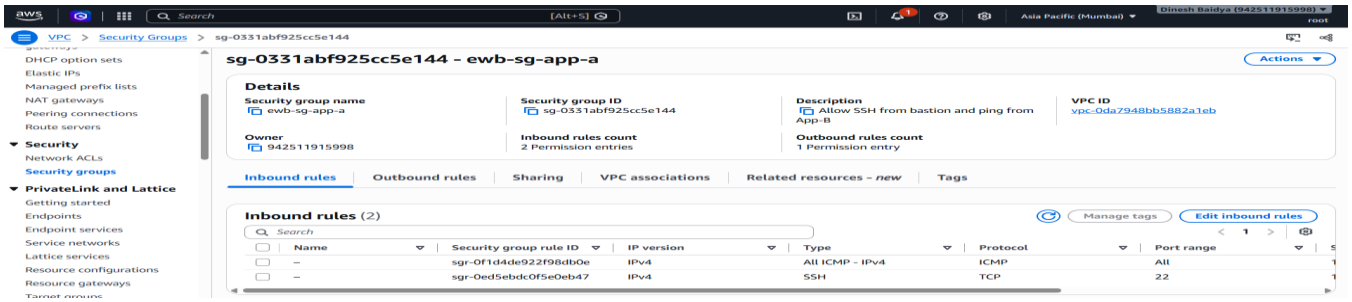
Security groups are stateful. The following describes the intended minimum inbound rules based on the lab configuration. Confirm the live rules and fill in any missing security-group IDs before submission.

Security group / target	Protocol / port	Source	Purpose
Bastion SG	TCP 22 (SSH)	Your current public IPv4 /32	Laptop to bastion; source must match current public IP
App-A SG	TCP 22 (SSH)	10.10.1.39/32	SSH from bastion if required by lab
App-A SG	ICMP Echo request	10.20.1.96/32	Permit App-B to ping App-A
ewb-sg-app-b	TCP 22 (SSH)	10.10.1.39/32	Permit SSH from bastion to App-B
ewb-sg-app-b	ICMP Echo request	10.10.2.192/32	Permit App-A to ping App-B

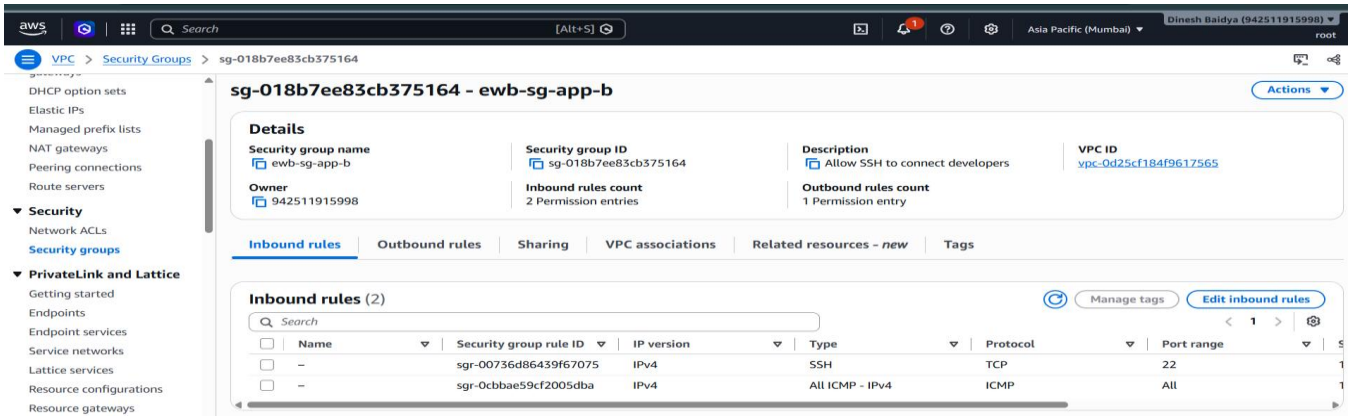
Inbound rules for the bastion security group



Inbound rules for the App-A security group



Inbound rules for ewb-sg-app-b



7. Verification and Test Results

The following results reflect the tests reported during the lab session. Insert your own terminal screenshots in the spaces provided. The initial baseline test must only be marked as evidenced if you actually captured it before peering/routes were configured.

Test	Command / location	Observed result	Evidence status
Forward ping	On App-A: ping -c 4 10.20.1.96	Reported success: 4/4 replies, 0% packet loss	Insert screenshot
Reverse ping	On App-B: ping -c 4 10.10.2.192	Reported success: 4/4 replies, 0% packet loss	Insert screenshot
Private SSH to App-B	Windows PowerShell: ssh ewb-app-b	Reported successful shell access through bastion ProxyJump	Insert screenshot
Private IP verification	On App-A and App-B: hostname; hostname -I	App-A: 10.10.2.192; App-B: 10.20.1.96; verify no public IPv4 in console	Insert screenshots
Baseline before peering	On App-A, before peering: ping -c 4 10.20.1.96	Expected no replies; mark only if actually captured at that stage	[Insert if captured; otherwise say not captured]
Controlled ICMP failure	Temporarily remove only App-B ICMP inbound rule from 10.10.2.192/32, then ping from App-A	User reported completing the controlled failure/recovery procedure; insert the actual failure output	Insert screenshot
ICMP recovery	Restore App-B ICMP inbound rule; repeat ping from App-A	Expected successful replies after rule restoration; include actual output	Insert screenshot

App-A forward ping to App-B: ping -c 4 10.20.1.96

```
PS C:\Users\ASUS> ssh ewb-app-a
       #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Sat Oct 31 14:33:52 2026 from 10.10.1.39
[ec2-user@ip-10-10-2-192 ~]$ ping -c 4 10.20.1.96
PING 10.20.1.96 (10.20.1.96) 56(84) bytes of data.
64 bytes from 10.20.1.96: icmp_seq=1 ttl=127 time=0.167 ms
64 bytes from 10.20.1.96: icmp_seq=2 ttl=127 time=0.215 ms
64 bytes from 10.20.1.96: icmp_seq=3 ttl=127 time=0.182 ms
64 bytes from 10.20.1.96: icmp_seq=4 ttl=127 time=0.219 ms

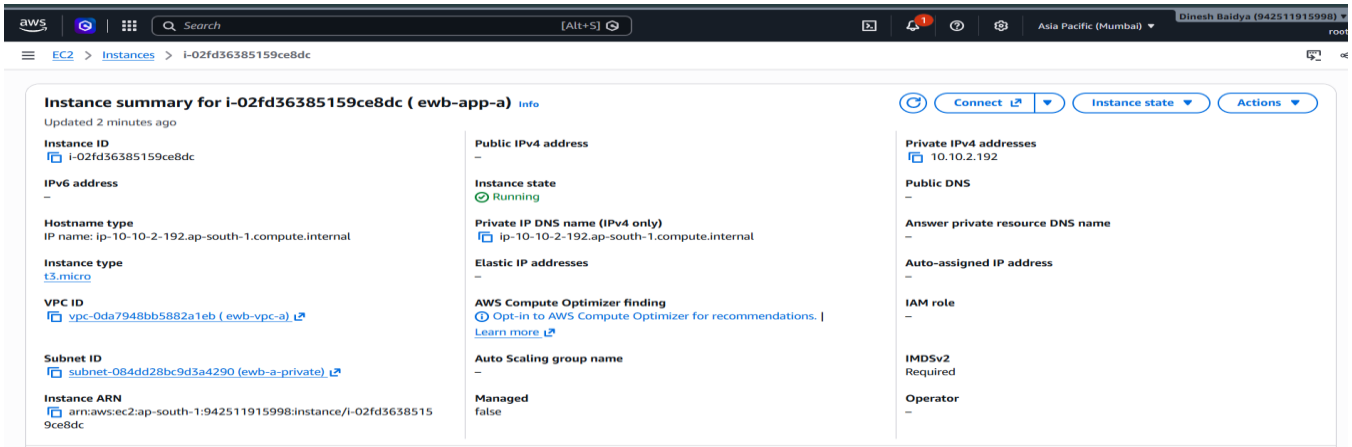
--- 10.20.1.96 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3133ms
rtt min/avg/max/mdev = 0.167/0.195/0.219/0.021 ms
[ec2-user@ip-10-10-2-192 ~]$ ping -c 4 10.20.1.96
PING 10.20.1.96 (10.20.1.96) 56(84) bytes of data.
```

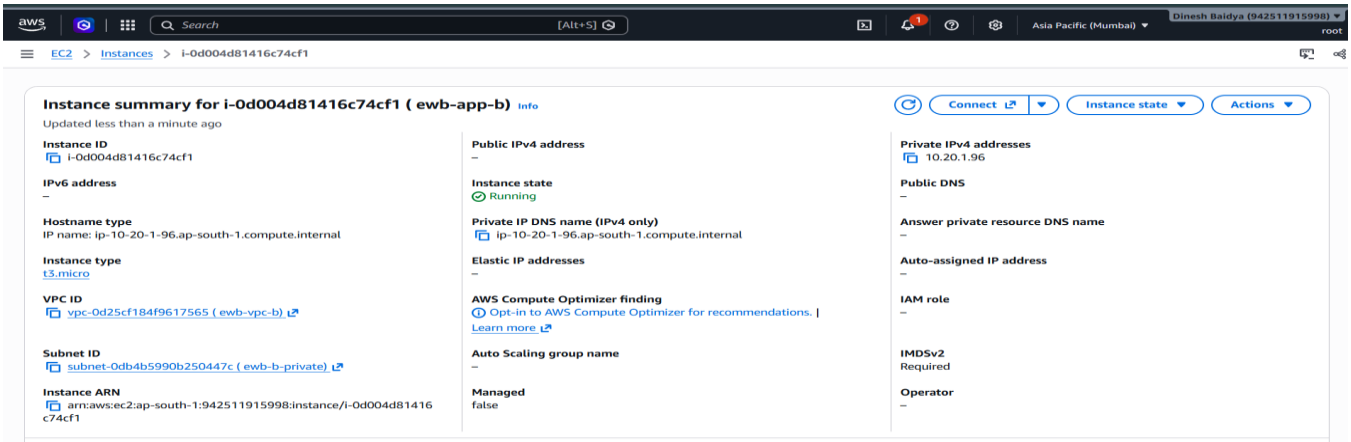
App-B reverse ping to App-A: ping -c 4 10.10.2.192

```
PS C:\Users\ASUS> ssh ewb-app-b
The authenticity of host '10.20.1.96 (<no hostip for proxy command>)' can't be established.
ED25519 key fingerprint is SHA256:QBaX0I5lOvv2vfFbPmkU8d4QqnNQwQRBOpZ9EtGqqII.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.20.1.96' (ED25519) to the list of known hosts.
       #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-20-1-96 ~]$ ping -c 4 10.10.2.192
PING 10.10.2.192 (10.10.2.192) 56(84) bytes of data.
64 bytes from 10.10.2.192: icmp_seq=1 ttl=127 time=0.168 ms
64 bytes from 10.10.2.192: icmp_seq=2 ttl=127 time=0.175 ms
64 bytes from 10.10.2.192: icmp_seq=3 ttl=127 time=0.180 ms
64 bytes from 10.10.2.192: icmp_seq=4 ttl=127 time=0.167 ms

--- 10.10.2.192 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3159ms
rtt min/avg/max/mdev = 0.167/0.172/0.180/0.005 ms
[ec2-user@ip-10-20-1-96 ~]$ hostname -I
10.20.1.96
[ec2-user@ip-10-20-1-96 ~]$ exit
logout
Connection to 10.20.1.96 closed.
```

EC2 console details showing App-A and App-B have no public IPv4





Controlled failure: forward ping fails while App-B ICMP rule is removed

```
PS C:\Users\ASUS> ssh ewb-app-a
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Sat Oct 31 14:44:40 2026 from 10.10.1.39
[ec2-user@ip-10-10-2-192 ~]$ ping -c 4 10.20.1.96
PING 10.20.1.96 (10.20.1.96) 56(84) bytes of data.

--- 10.20.1.96 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3095ms

[ec2-user@ip-10-10-2-192 ~]$ ping -c 4 10.20.1.96
```

Recovery: forward ping succeeds after the ICMP rule is restored

```
[ec2-user@ip-10-10-2-192 ~]$ ping -c 4 10.20.1.96
PING 10.20.1.96 (10.20.1.96) 56(84) bytes of data.
64 bytes from 10.20.1.96: icmp_seq=1 ttl=127 time=0.159 ms
64 bytes from 10.20.1.96: icmp_seq=2 ttl=127 time=0.164 ms
64 bytes from 10.20.1.96: icmp_seq=3 ttl=127 time=0.242 ms
64 bytes from 10.20.1.96: icmp_seq=4 ttl=127 time=0.183 ms

--- 10.20.1.96 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3107ms
rtt min/avg/max/mdev = 0.159/0.187/0.242/0.033 ms
[ec2-user@ip-10-10-2-192 ~]$
```

8. Traffic Flow Explanation

Forward path: App-A to App-B

App-A (10.10.2.192) sends ICMP traffic to App-B (10.20.1.96). The ewb-rt-a-private route table matches destination 10.20.0.0/16 and forwards the traffic to the VPC peering connection pcx-0e2a0f1d635f24a36. The peering connection carries the traffic to VPC B. App-B's security group must allow ICMP Echo requests from 10.10.2.192/32.

Return path: App-B to App-A

Replies travel from App-B toward 10.10.2.192. The ewb-rt-b-private route table matches destination 10.10.0.0/16 and sends traffic to the same peering connection. App-A's security group must permit the inbound ICMP request from App-B. Both route directions and the destination security group rules are required for successful communication.

SSH path to App-B

The laptop opens SSH to the public bastion. The bastion then initiates SSH to App-B's private address 10.20.1.96 using ProxyJump. The public subnet route table has a specific route for 10.20.0.0/16 to the peering connection, while its 0.0.0.0/0 route points to the Internet Gateway. App-B allows TCP 22 only from the bastion private IP 10.10.1.39/32. App-B does not need a public IPv4 address.

9. Controlled Failure and Recovery

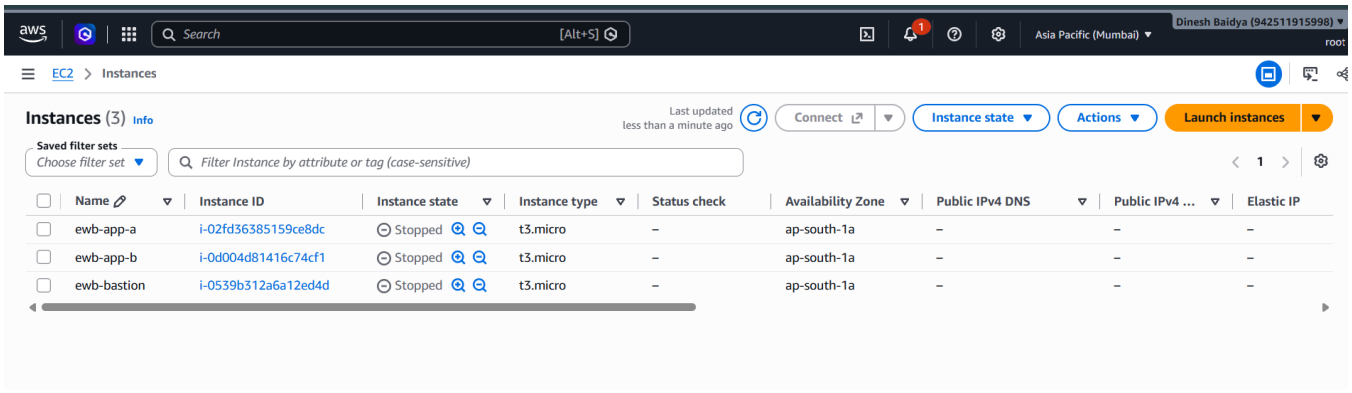
For the controlled test, only the App-B inbound ICMP Echo request rule sourced from App-A (10.10.2.192/32) is temporarily removed. The forward ping should fail while that rule is absent. Restoring the same rule should allow the ping to succeed again. The SSH rule is kept in place throughout. This demonstrates that routing alone does not override security-group filtering. Attach screenshots showing both the failure and recovery.

10. Cleanup and Cost Control

After saving all evidence and the final report, removed resources created for this lab.

1. Terminate ewb-bastion, ewb-app-a and ewb-app-b.
2. Confirmed lab EBS volumes are deleted; remove leftover lab volumes or snapshots if appropriate.
3. Released any Elastic IP allocated for this lab.
4. Deleted the VPC peering connection and remove its routes from the lab route tables.
5. After network interfaces disappear, delete lab security groups.
6. Removed subnet associations as required; delete lab subnets and custom route tables.
7. Detach and delete ewb-igw-a, then delete the two lab VPCs.
8. Check EC2, VPC, EBS, public IPv4 allocations and Billing in ap-south-1.

Cleanup evidence: lab instances terminated



11. Learning Outcomes

This lab reinforced that VPC peering provides private connectivity but does not automatically add routes or allow traffic through security groups. Each participating subnet needs a correct route in the appropriate direction, and the return path is as important as the forward path. A bastion host can provide controlled SSH access to private instances without assigning those instances public IP addresses. A controlled security-group change is a useful way to distinguish a routing problem from a traffic-filtering problem.

12. Final Submission Checklist

- Cover page includes name, batch, project title, Region and completion date.
- Architecture diagram and actual resource inventory IDs are included.
- Subnet CIDRs and all three route tables with subnet associations are evidenced.
- Peering connection state and inbound rules for each security group are shown.
- Forward ping, reverse ping and SSH through bastion screenshots are included.
- Controlled ICMP failure and recovery are evidenced.
- Baseline failure is included only if actually captured before peering.
- Cleanup evidence and personal learning note are included.
- No private keys, passwords, access tokens or other secrets are included.